

1 Semi-implicit

$$\begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\frac{cFr}{2}\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & -\frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix}$$

this implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & -\frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \left[\begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\frac{cFr}{2}\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right] + \begin{pmatrix} 0 \\ -\frac{c}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ -\frac{c}{2}\Delta tk(1 - \frac{cFr}{2}\Delta tk) & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\frac{cFr}{2}\Delta tk & 0 \\ \frac{c^2Fr}{4}\Delta t^2 k^2 - \frac{c}{2}\Delta tk & -\frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right]$$

2 Segregated Implicit Advection, Implicit Gravity

$$\begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & -\frac{c}{2}\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\frac{cFr}{2}\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_3 \\ h_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_3 \\ h_3 \end{pmatrix}$$

this implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \right]$$

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\quad \left[\begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \left[\begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\frac{cFr}{2}\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} \right] \right] \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\quad \left[\begin{pmatrix} 1 & 0 \\ \frac{c^2 Fr}{4}\Delta t^2 k^2 - c\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \frac{c^2 Fr}{4}\Delta t^2 k^2 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} \right] \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\quad \left[\begin{pmatrix} 1 & 0 \\ \frac{c^2 Fr}{4}\Delta t^2 k^2 - c\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \frac{c^2 Fr}{4}\Delta t^2 k^2 & 0 \end{pmatrix} \begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} \right] \end{aligned}$$

3 Operator Split Implicit Advection (both) and Implicit Gravity

$$\begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} + \begin{pmatrix} 0 & -\frac{c}{2}\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

which implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right] + \begin{pmatrix} 0 & -\frac{c}{2}\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \frac{cFr}{2}\Delta tk - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \right] \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

no dependence on previous iteration in this case - so doesn't improve with more iterations

4 Operator Split Implicit Momentum Advection, Explicit Mass Advection and Implicit Gravity

$$\begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -\frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} + \begin{pmatrix} 0 & -\frac{c}{2}\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

which implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -\frac{cFr}{2}\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right] +$$

$$\begin{aligned}
\begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\frac{c}{2}\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix}^{-1} \\
&\quad \left[\left\{ \begin{pmatrix} 0 & 0 \\ -\frac{c}{2}\Delta tk & 0 \end{pmatrix} \begin{pmatrix} 1 + \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 - \frac{cFr}{2}\Delta tk & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -\frac{c}{2}\Delta tk & 1 - \frac{cFr}{2}\Delta tk + \frac{c^2}{4}\Delta t^2 k^2 \end{pmatrix} \right\} \right. \\
&\quad \left. + \begin{pmatrix} 0 & -\frac{c}{2}\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right]
\end{aligned}$$

now there is dependence on previous iterations